Note of Schwinger boson

(Dated: October 2, 2026)

SCHWINGER BOSON REPRESENTATION OF SPIN OPERATORS

Assume that the Hamiltonian takes the form [1, 2]

$$H = \sum_b J_b \mathbf{S}_\alpha \cdot \mathbf{S}_j. \tag{1}$$

For each bond, the spin operators can be represented in the Schwinger boson formalism as

$$\begin{aligned}
\mathbf{S}_\alpha \cdot \mathbf{S}_j &=: B^\dagger_{\alpha\gamma} B_{\alpha\gamma} : -A^\dagger_{\alpha\gamma} A_{\alpha\gamma}, \\
B_{\alpha\gamma} &= \frac{1}{2}(b_{\alpha\uparrow} b^\dagger_{\gamma\uparrow} + b_{\alpha\downarrow} b^\dagger_{\gamma\downarrow}), \\
B^\dagger_{\alpha\gamma} &= \frac{1}{2}(b^\dagger_{\alpha\uparrow} b_{\gamma\uparrow} + b^\dagger_{\alpha\downarrow} b_{\gamma\downarrow}), \\
A_{\alpha\gamma} &= \frac{1}{2}(b_{\alpha\uparrow} b_{\gamma\downarrow} - b_{\alpha\downarrow} b_{\gamma\uparrow}), \\
A^\dagger_{\alpha\gamma} &= \frac{1}{2}(b^\dagger_{\alpha\uparrow} b^\dagger_{\gamma\downarrow} - b^\dagger_{\alpha\downarrow} b^\dagger_{\gamma\uparrow}),
\end{aligned} \tag{2}$$

where $b_{\alpha\sigma}$ is the Schwinger boson operator and :: denotes normal ordering. In the path-integral formalism, we then write

$$\begin{aligned}
Z &= \int \mathcal{D}[\bar{b}, b]\, \mathcal{D}[\lambda]\, e^{-S}, \\
S &= \int_0^\beta d\tau \Big[\sum_\alpha \bar{b}_{\alpha\sigma} \partial_\tau b_{\alpha\sigma} + \mathcal{H}(\bar{b}, b) + \mathcal{J}_s \\
&\quad + \mathcal{J}_b + i \sum_\alpha \lambda_\alpha (\bar{b}_{\alpha\sigma} b_{\alpha\sigma} - 2S) \Big], \\
\mathcal{J}_s &= \sum_\alpha j^{\tau\mu}_\alpha \mathbf{b}^{\tau\dagger}_\alpha \cdot \sigma^\mu \cdot \mathbf{b}^\tau_\alpha, \\
\mathcal{J}_b &= \sum_\alpha h S^x_\alpha,
\end{aligned} \tag{3}$$

Using the Hubbard-Stratonovich transformation, we decouple the quartic term in the Hamiltonian and obtain the mean-field Hamiltonian:

$$\begin{aligned}
e^{J_b A^\dagger_{\alpha\gamma} A_{\alpha\gamma}} &= \int \mathcal{D}[\bar{W}^A_b, W^A_b]\, e^{-J_b \bar{W}^A_b W^A_b + J_b(-\bar{W}^A_b A_{\alpha\gamma} + W^A_b A^\dagger_{\alpha\gamma})}, \\
e^{J_b : B^\dagger_{\alpha\gamma} B_{\alpha\gamma} :} &= \int \mathcal{D}[\bar{W}^B_b, W^B_b]\, e^{-J_b \bar{W}^B_b W^B_b + J_b(\bar{W}^B_b B_{\alpha\gamma} + W^B_b B^\dagger_{\alpha\gamma})}.
\end{aligned} \tag{4}$$

Note that the signs in Eq. 4 differ for $A_{\alpha\gamma}$ and $B_{\alpha\gamma}$ because these terms enter the Hamiltonian with opposite signs. Expanding Eq. 3 around the saddle point, we obtain

$$\begin{aligned}
S_{\text{eff}} &= S^{\text{sp}}_{\text{eff}} + \sum_{\alpha_1, \alpha_2} S^2_{\alpha_1, \alpha_2} \Delta_{\alpha_1} \Delta_{\alpha_2} + S_{\text{int}}, \\
S^{\text{sp}}_{\text{eff}} &= -N_s \beta \sum_{\delta > 0} J_\delta \left(B^2_\delta - A^2_\delta \right) - 2S N_s \beta \lambda - \ln \mathcal{Z}^{\text{sp}}_{\text{bos}}, \\
\mathcal{Z}^{\text{sp}}_{\text{bos}} &= \int \mathcal{D}[\bar{b}, b]\, e^{-S^{\text{sp}}_{\text{bos}}}, \\
S^{\text{sp}}_{\text{bos}} &= \sum_{k, \omega_n} b^\dagger_{k, \omega_n} M_{k, \omega_n} b_{k, \omega_n}
\end{aligned} \tag{5}$$

At the saddle point, the self-consistency equations are[3]

$$\begin{aligned}
\left. \frac{\partial S_{\text{eff}}}{\partial \phi_\alpha} \right|_{\text{sp}} &= \left. \frac{\partial S_0}{\partial \phi_\alpha} \right|_{\text{sp}} + \frac{1}{2} \operatorname{Tr} \left[\mathcal{G}^{\text{sp}} \frac{\partial \mathcal{G}^{-1}}{\partial \phi_\alpha} \right] = 0, \\
\mathcal{G}^{-1} &= M.
\end{aligned} \tag{6}$$

The mean-field parameters are given by

$$\begin{aligned}
W^A_b &= i A_\delta, \\
W^B_b &= B_\delta, \\
\lambda &= \lambda.
\end{aligned} \tag{7}$$

In the momentum basis $\psi = \begin{pmatrix} b^\dagger_{k,\sigma} \\ b_{-k,\sigma} \end{pmatrix}$, the M matrix is defined by

$$\begin{aligned}
\psi^\dagger M \psi &= \begin{pmatrix} b_{k,\sigma} & b^\dagger_{-k,\sigma} \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} b^\dagger_{k,\sigma} \\ b_{-k,\sigma} \end{pmatrix} \\
&= A b_{k,\sigma} b^\dagger_{k,\sigma} + B b_{k,\sigma} b_{-k,\sigma} + C b^\dagger_{-k,\sigma} b^\dagger_{k,\sigma} + D b^\dagger_{-k,\sigma} b_{-k,\sigma}.
\end{aligned} \tag{8}$$

Thus, because of the sign difference in Eq. 4, each bond in Eq. 1 contributes the following operator terms to M in the momentum basis:

$$\begin{aligned}
& b^\dagger_{\alpha,-k,\sigma} b_{\gamma,-k,\sigma}, \quad && b_{\alpha,k,\sigma} b^\dagger_{\gamma,k,\sigma}, \\
& b_{\alpha,k,\uparrow} b_{\gamma,-k,\downarrow}, \quad && -b_{\alpha,k,\downarrow} b_{\gamma,-k,\uparrow}, \\
& b^\dagger_{\alpha,-k,\uparrow} b^\dagger_{\gamma,k,\downarrow}, \quad && -b^\dagger_{\alpha,-k,\downarrow} b^\dagger_{\gamma,k,\uparrow}.
\end{aligned} \tag{9}$$

Here, we assume the Fourier transform

$$b_\alpha = \frac{1}{N} \sum_k b_k e^{ik \cdot r_\alpha}, \qquad b^\dagger_\alpha = \frac{1}{N} \sum_k b^\dagger_k e^{-ik \cdot r_\alpha}, \tag{10}$$

without imposing any a priori order. This gives

$$\begin{aligned}
B_{\alpha\gamma} + B^\dagger_{\alpha\gamma} &= \frac{1}{2} \left(b_{\alpha\uparrow} b^\dagger_{\gamma\uparrow} + b_{\alpha\downarrow} b^\dagger_{\gamma\downarrow} \right) + \frac{1}{2} \left(b^\dagger_{\alpha\uparrow} b_{\gamma\uparrow} + b^\dagger_{\alpha\downarrow} b_{\gamma\downarrow} \right) \\
&= \frac{1}{2} \sum_k \Big[b_{\alpha,k,\uparrow} b^\dagger_{\gamma,k,\uparrow} e^{ik \cdot r_{\alpha\gamma}} + b_{\alpha,k,\downarrow} b^\dagger_{\gamma,k,\downarrow} e^{ik \cdot r_{\alpha\gamma}} \\
&\quad + b^\dagger_{\alpha,-k,\uparrow} b_{\gamma,-k,\uparrow} e^{-ik \cdot r_{\alpha\gamma}} + b^\dagger_{\alpha,-k,\downarrow} b_{\gamma,-k,\downarrow} e^{-ik \cdot r_{\alpha\gamma}} \Big].
\end{aligned} \tag{11}$$

with $r_{\alpha\gamma} = r_j - r_\alpha$. Considering the bond (i, j) and the bond (j, i), we have $r_{\gamma\alpha} = -r_{\alpha\gamma}$. For the bond (i, j),

$$\begin{aligned}
B_{\alpha\gamma} + B^\dagger_{\alpha\gamma} &= \frac{1}{2} \sum_k \Big[b_{\alpha,k,\uparrow} b^\dagger_{\gamma,k,\uparrow} e^{ik \cdot r_{\alpha\gamma}} + b_{\alpha,k,\downarrow} b^\dagger_{\gamma,k,\downarrow} e^{ik \cdot r_{\alpha\gamma}} \\
&\quad + b^\dagger_{\alpha,-k,\uparrow} b_{\gamma,-k,\uparrow} e^{ik \cdot r_{\alpha\gamma}} + b^\dagger_{\alpha,-k,\downarrow} b_{\gamma,-k,\downarrow} e^{ik \cdot r_{\alpha\gamma}} \Big],
\end{aligned} \tag{12}$$

where we label these terms as ①–④. For the bond (j, i),

$$\begin{aligned}
B_{\gamma\alpha} + B^\dagger_{\gamma\alpha} &= \frac{1}{2} \sum_k \Big[b_{\gamma,k,\uparrow} b^\dagger_{\alpha,k,\uparrow} e^{-ik \cdot r_{\alpha\gamma}} + b_{\gamma,k,\downarrow} b^\dagger_{\alpha,k,\downarrow} e^{-ik \cdot r_{\alpha\gamma}} \\
&\quad + b^\dagger_{\gamma,-k,\uparrow} b_{\alpha,-k,\uparrow} e^{-ik \cdot r_{\alpha\gamma}} + b^\dagger_{\gamma,-k,\downarrow} b_{\alpha,-k,\downarrow} e^{-ik \cdot r_{\alpha\gamma}} \Big],
\end{aligned} \tag{13}$$

where we label these terms as ⑤–⑧. Combining the contributions from the two bonds, we obtain

$$\begin{aligned} ① + ⑤ &\to \cos k, \quad ② + ⑥ \to \cos k, \\ ③ + ⑦ &\to \cos k, \quad ④ + ⑧ \to \cos k. \end{aligned} \tag{14}$$

For the A term, we start from the definitions

$$\begin{aligned} A_{\alpha\gamma} &= \frac{1}{2}\left(b_{\alpha\uparrow}b_{\gamma\downarrow} - b_{\alpha\downarrow}b_{\gamma\uparrow}\right), \\ A^{\dagger}_{\alpha\gamma} &= \frac{1}{2}\left(b^{\dagger}_{\alpha\uparrow}b^{\dagger}_{\gamma\downarrow} - b^{\dagger}_{\alpha\downarrow}b^{\dagger}_{\gamma\uparrow}\right). \end{aligned} \tag{15}$$

If $W_{A_{\alpha\gamma}}$ is real, we use the combinations

$$A_{\alpha\gamma} + A^{\dagger}_{\alpha\gamma}, \qquad A_{\gamma\alpha} + A^{\dagger}_{\gamma\alpha}. \tag{16}$$

If $W_{A_{\alpha\gamma}}$ is imaginary, we use the combinations involving the imaginary unit i:

$$i\left(A_{\alpha\gamma} - A^{\dagger}_{\alpha\gamma}\right), \qquad i\left(-A_{\gamma\alpha} + A^{\dagger}_{\gamma\alpha}\right). \tag{17}$$

Using the Fourier transform in Eq. 10, this reduces to

$$\begin{aligned} A_{\alpha\gamma} - A^{\dagger}_{\alpha\gamma} = \frac{1}{2}\sum_{k}\Big[&b_{\alpha,k,\uparrow}b_{\gamma,-k,\downarrow}e^{ik\cdot r_{\alpha\gamma}} - b_{\alpha,k,\downarrow}b_{\gamma,-k,\uparrow}e^{ik\cdot r_{\alpha\gamma}} \\ &- b^{\dagger}_{\alpha,k,\uparrow}b^{\dagger}_{\gamma,-k,\downarrow}e^{-ik\cdot r_{\alpha\gamma}} + b^{\dagger}_{\alpha,k,\downarrow}b^{\dagger}_{\gamma,-k,\uparrow}e^{-ik\cdot r_{\alpha\gamma}}\Big]. \end{aligned} \tag{18}$$

For the bond (i, j), this gives

$$\begin{aligned} A_{\alpha\gamma} - A^{\dagger}_{\alpha\gamma} = \frac{1}{2}\sum_{k}\Big[&b_{\alpha,k,\uparrow}b_{\gamma,-k,\downarrow}e^{ik\cdot r_{\alpha\gamma}} - b_{\alpha,k,\downarrow}b_{\gamma,-k,\uparrow}e^{ik\cdot r_{\alpha\gamma}} \\ &- b^{\dagger}_{\alpha,k,\uparrow}b^{\dagger}_{\gamma,-k,\downarrow}e^{-ik\cdot r_{\alpha\gamma}} + b^{\dagger}_{\alpha,k,\downarrow}b^{\dagger}_{\gamma,-k,\uparrow}e^{-ik\cdot r_{\alpha\gamma}}\Big]. \end{aligned} \tag{19}$$

where we label these terms as ①–④. For the bond (j, i), using $r_{\gamma\alpha} = -r_{\alpha\gamma}$, we have

$$\begin{aligned} -A_{\gamma\alpha} + A^{\dagger}_{\gamma\alpha} = \frac{1}{2}\sum_{k}\Big[&b_{\gamma,k,\uparrow}b_{\alpha,-k,\downarrow}e^{-ik\cdot r_{\alpha\gamma}} - b_{\gamma,k,\downarrow}b_{\alpha,-k,\uparrow}e^{-ik\cdot r_{\alpha\gamma}} \\ &- b^{\dagger}_{\gamma,k,\uparrow}b^{\dagger}_{\alpha,-k,\downarrow}e^{ik\cdot r_{\alpha\gamma}} + b^{\dagger}_{\gamma,k,\downarrow}b^{\dagger}_{\alpha,-k,\uparrow}e^{ik\cdot r_{\alpha\gamma}}\Big], \end{aligned} \tag{20}$$

where we label these terms as ⑤–⑧. Therefore, we have

$$\begin{aligned} ①^{\dagger} &= ③, \quad ②^{\dagger} = ④, \\ ⑤^{\dagger} &= ⑦, \quad ⑥^{\dagger} = ⑧. \end{aligned} \tag{21}$$

Combining the contributions from the two bonds, we obtain

$$\begin{aligned} ① + ⑤ &\to \sin(k\cdot r_{\alpha\gamma}), \quad ② + ⑥ \to -\sin(k\cdot r_{\alpha\gamma}), \\ ③ + ⑦ &\to \sin(k\cdot r_{\alpha\gamma}), \quad ④ + ⑧ \to -\sin(k\cdot r_{\alpha\gamma}). \end{aligned} \tag{22}$$

Thus, the A term reduces to a combination of $\pm\sin(k\cdot r_{\alpha\gamma})$ contributions.

Therefore, by rewriting

$$\mathbf{S}_{\alpha}\cdot\mathbf{S}_{j} = \frac{1}{2}\left(\mathbf{S}_{\alpha}\cdot\mathbf{S}_{j} + \mathbf{S}_{j}\cdot\mathbf{S}_{\alpha}\right), \tag{23}$$

we can transform the Schwinger boson representation into a combination of cos and sin terms. Note that we do not assume any a priori order in this process. When applying this to the triangular lattice, we assume a 120° ordered state, with a magnetic unit cell containing three sites. We then list all bonds within and between magnetic unit cells and calculate each bond's contribution to M. Finally, we solve the self-consistency equations to obtain the mean-field parameters and the spinon spectrum.

PRIORI ORDER

A PRIORI ORDER

Although the Schwinger boson mean-field theory does not require any *a priori* order, it is helpful to add a priori order to the system to ensure that the solution converges to the correct saddle point. Following Refs. [4–6], we add a priori 120° order to the system, which is a spiral phase $\mathbf{S} = (S\cos Q, S\sin Q, 0)$. Assuming there is magnetic order in a plane in the classical limit ($S \to \infty$), we have

$$\begin{aligned} \mathbf{S}_{\alpha}\cdot\mathbf{S}_{\gamma} &= S^2\cos(\Delta\theta) \\ &= |B_{\alpha\gamma}|^2 - |A_{\alpha\gamma}|^2. \end{aligned} \tag{24}$$

Then, we can set $B_{\alpha\gamma} \sim S\cos(\Delta\theta/2)$ and $A_{\alpha\gamma} \sim S\sin(\Delta\theta/2)$ as the initial values of the mean-field parameters. Therefore, with the magnetic unit cell $\{\alpha\}$, the Fourier transform of the Bose operators becomes

$$b_{\alpha\sigma} = \frac{1}{N}\sum_{k} b_{k\sigma}e^{i(k+\sigma Q/2)\cdot r_{\alpha}}, \tag{25}$$

where the spin-dependent shift is $k \to k + \sigma Q/2$.

However, for the triangle $\{r_{01}, r_{02}, r'_{02}\} = \{(1,0), (1/2, \sqrt{3}/2), (-1/2, \sqrt{3}/2)\}$, it forms the set of equations

$$\begin{cases} q_1 + 0 = \pi/3 + 2k\pi, \\ -q_1/2 + \sqrt{3}/2\, q_2 = -\pi/3 + 2n\pi, \\ q_1/2 + \sqrt{3}/2\, q_2 = \pi/3 + 2m\pi. \end{cases} \tag{26}$$

Its solution is given by $(q_1, q_2) = (\pi/3, \pi/\sqrt{3})$ and $(k, n, m) = (0, 1/2, 0)$.

Therefore, with $Q = (q_1, q_2)$ and an additional phase π for r_{01}, Eqs. 12 and 13 become

$$\begin{aligned} B_{\alpha\gamma} + B^{\dagger}_{\alpha\gamma} = \frac{1}{2}\sum_{k}\Big[&b_{\alpha,k,\uparrow}b^{\dagger}_{\gamma,k,\uparrow}e^{i(k+Q/2)\cdot r_{\alpha\gamma}} + b_{\alpha,k,\downarrow}b^{\dagger}_{\gamma,k,\downarrow}e^{i(k-Q/2)\cdot r_{\alpha\gamma}} \\ &+ b^{\dagger}_{\alpha,-k,\uparrow}b_{\gamma,-k,\uparrow}e^{i(k-Q/2)\cdot r_{\alpha\gamma}} + b^{\dagger}_{\alpha,-k,\downarrow}b_{\gamma,-k,\downarrow}e^{i(k+Q/2)\cdot r_{\alpha\gamma}}\Big], \end{aligned} \tag{27}$$

$$\begin{aligned} B_{\gamma\alpha} + B^{\dagger}_{\gamma\alpha} = \frac{1}{2}\sum_{k}\Big[&b_{\alpha,k,\uparrow}b^{\dagger}_{\gamma,k,\uparrow}e^{-i(k+Q/2)\cdot r_{\alpha\gamma}} + b_{\alpha,k,\downarrow}b^{\dagger}_{\gamma,k,\downarrow}e^{-i(k-Q/2)\cdot r_{\alpha\gamma}} \\ &+ b^{\dagger}_{\alpha,-k,\uparrow}b_{\gamma,-k,\uparrow}e^{-i(k-Q/2)\cdot r_{\alpha\gamma}} + b^{\dagger}_{\alpha,-k,\downarrow}b_{\gamma,-k,\downarrow}e^{-i(k+Q/2)\cdot r_{\alpha\gamma}}\Big], \end{aligned} \tag{28}$$

$$
\begin{aligned}
A_{\alpha\gamma} - A^{\dagger}_{\alpha\gamma} = \frac{1}{2}\sum_{k}\Big[& b_{\alpha,k,\uparrow} b_{\gamma,-k,\downarrow} e^{i(k+Q/2)\cdot r_{\alpha\gamma}} \\
& - b_{\alpha,k,\downarrow} b_{\gamma,-k,\uparrow} e^{i(k-Q/2)\cdot r_{\alpha\gamma}} \\
& - b^{\dagger}_{\alpha,k,\uparrow} b^{\dagger}_{\gamma,-k,\downarrow} e^{-i(k+Q/2)\cdot r_{\alpha\gamma}} \\
& + b^{\dagger}_{\alpha,k,\downarrow} b^{\dagger}_{\gamma,-k,\uparrow} e^{-i(k-Q/2)\cdot r_{\alpha\gamma}} \Big].
\end{aligned} \tag{29}
$$

where we label these terms as ①–④. For the bond (γ, α), using $r_{\gamma\alpha} = -r_{\alpha\gamma}$, we have

$$
\begin{aligned}
-A_{\gamma\alpha} + A^{\dagger}_{\gamma\alpha} = \frac{i}{2}\sum_{k}\Big[& b_{\gamma,k,\uparrow} b_{\alpha,-k,\downarrow} e^{-i(k+Q/2)\cdot r_{\alpha\gamma}} \\
& - b_{\gamma,k,\downarrow} b_{\alpha,-k,\uparrow} e^{-i(k-Q/2)\cdot r_{\alpha\gamma}} \\
& - b^{\dagger}_{\gamma,k,\uparrow} b^{\dagger}_{\alpha,-k,\downarrow} e^{-i(k+Q/2)\cdot r_{\alpha\gamma}} \\
& + b^{\dagger}_{\gamma,k,\downarrow} b^{\dagger}_{\alpha,-k,\uparrow} e^{-i(k-Q/2)\cdot r_{\alpha\gamma}} \Big],
\end{aligned} \tag{30}
$$

[1] E. A. Ghioldi, M. G. Gonzalez, S.-S. Zhang, Y. Kamiya, L. O. Manuel, A. E. Trumper, and C. D. Batista, Phys. Rev. B **98**, 184403 (2018).

[2] I. L. Pomponio, E. A. Ghioldi, C. J. Gazza, L. O. Manuel, and A. E. Trumper, Phys. Rev. B **110**, 224402 (2024).

[3] S. Toth and B. Lake, Journal of Physics: Condensed Matter **27**, 166002 (2015).

[4] A. E. Trumper, L. O. Manuel, C. J. Gazza, and H. A. Ceccatto, Phys. Rev. Lett. **78**, 2216 (1997).

[5] H. A. Ceccatto, C. J. Gazza, and A. E. Trumper, Phys. Rev. B **47**, 12329(R) (1993).

[6] C. Gazza and H. Ceccatto, Journal of Physics: Condensed Matter **5**, L135 (1993).